

SLAM – Simultaneous Localization and Mapping

Autonomous Navigation of Mobile Robots

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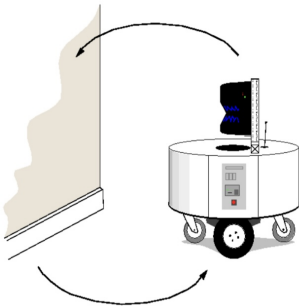
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What is SLAM?

- Estimate the pose of a robot and the map of the environment at the same time
- SLAM is hard, because
 - a map is needed for localization and
 - a good pose estimate is needed for mapping
- **Localization** – inferring location given a map
- **Mapping** – inferring a map given locations
- **SLAM** – learning a map and locating the robot simultaneously

The SLAM Problem

- SLAM has long been regarded as a chicken-or-egg problem
 - a map is needed for localization and
 - a pose estimate is needed for mapping



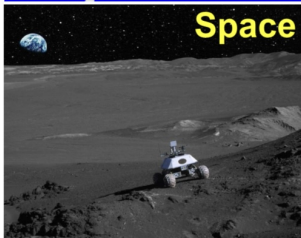
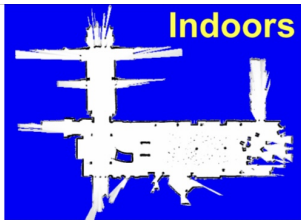
SLAM Applications

- SLAM is central to a range of indoor, outdoor, in-air and underwater applications for both manned and autonomous vehicles

Examples

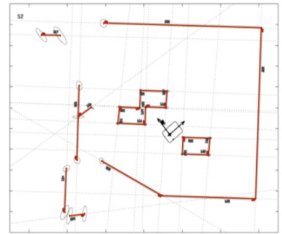
- At home – vacuum cleaner, lawn mower
- Air – surveillance with unmanned air vehicles
- Underwater – reef monitoring
- Underground – exploration of mines
- Space – terrain mapping for localization
- Every application that requires a map

SLAM Applications



Map Representations

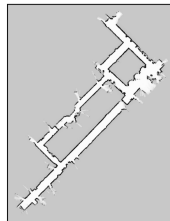
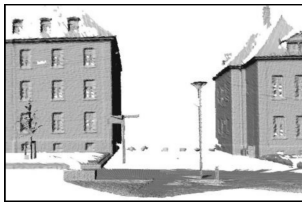
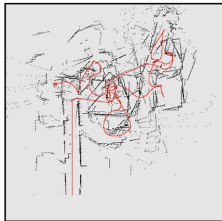
- Examples — Subway map, city map, landmark-based map



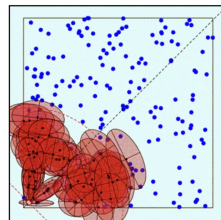
- Maps are topological and / or metric models of the environment

Map Representations in Robotics

- Grid maps or scans, 2D, 3D



- Landmark-based



The SLAM Problem

- SLAM is considered a fundamental problem for robots to become truly autonomous
- Large variety of different SLAM approaches have been developed
- The majority uses probabilistic concepts
- History of SLAM dates back to the mid-eighties

Feature-Based SLAM

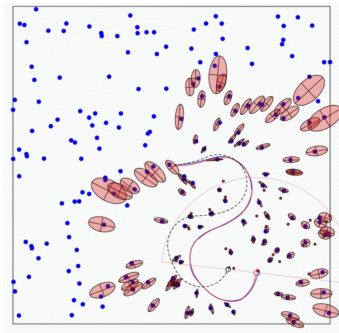
Given

- The robot's controls
 $U_{1:k} = \{u_1, u_2, \dots, u_k\}$

- Relative observations
 $Z_{1:k} = \{z_1, z_2, \dots, z_k\}$

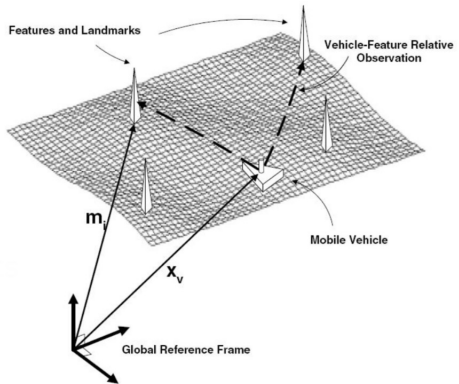
Wanted

- Map of features
 $m = \{m_1, m_2, \dots, m_n\}$
- Path of the robot
 $X_{1:k} = \{x_1, x_2, \dots, x_k\}$



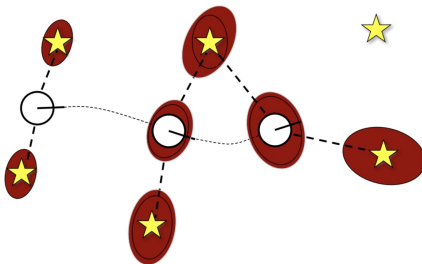
Feature-Based SLAM

- Absolute robot poses
- Absolute landmark positions
- But only relative measurements of landmarks



Why is SLAM a Hard Problem?

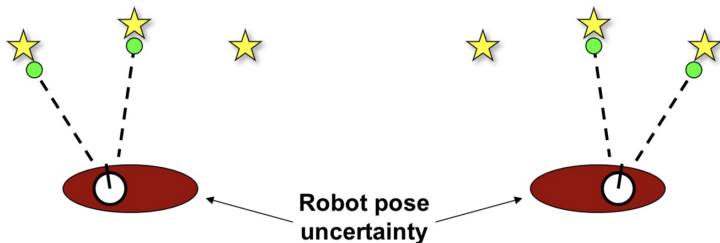
1. Robot path and map are both **unknown**



2. Errors in map and pose estimates correlated

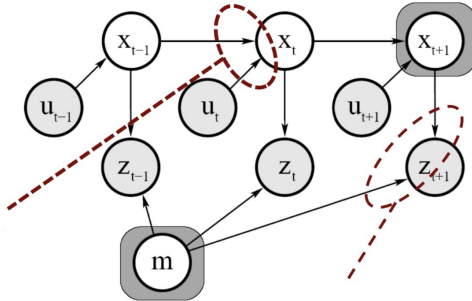
Why is SLAM a Hard Problem?

- The mapping between observations and landmarks is unknown
- Picking **wrong** data associations can have **catastrophic** consequences (divergence)



Motion and Observation Model

$x_t = f(x_{t-1}, u_t)$
“Motion model”



$z_t = h(x_t, m)$
“Observation model”

Remember the KF Algorithm

Algorithm `kalman_filter` ($\mu_{t-1}, \Sigma_{t-1}, u_t, z_t$) :

Prediction

$$\bar{\mu}_t = A_t \mu_{t-1} + B_t u_t$$

$$\bar{\Sigma}_t = A_t \Sigma_{t-1} A_t^T + Q_t$$

Correction

$$K_t = \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + R_t)^{-1}$$

$$\mu_t = \bar{\mu}_t + K_t (z_t - C_t \bar{\mu}_t)$$

$$\Sigma_t = (I - K_t C_t) \bar{\Sigma}_t$$

Return μ_t, Σ_t

EKF SLAM

State representation

■ Localization

3×1 pose vector

3×3 cov. matrix

$$x_k = \begin{bmatrix} x_k \\ y_k \\ \theta_k \end{bmatrix} \quad \Sigma_k = \begin{bmatrix} \sigma_x^2 & \sigma_{xy}^2 & \sigma_{x\theta}^2 \\ \sigma_{yx}^2 & \sigma_y^2 & \sigma_{y\theta}^2 \\ \sigma_{\theta x}^2 & \sigma_{\theta y}^2 & \sigma_\theta^2 \end{bmatrix}$$

■ SLAM

- Landmarks **simply extend** the state
- **Growing** state vector and covariance matrix!

$$x_k = \begin{bmatrix} x_R \\ m_1 \\ m_2 \\ \vdots \\ m_n \end{bmatrix} \quad \Sigma_k = \begin{bmatrix} \Sigma_R & \Sigma_{RM_1} & \Sigma_{RM_2} & \cdots & \Sigma_{RM_n} \\ \Sigma_{M_1R} & \Sigma_{M_1} & \Sigma_{M_1M_2} & \cdots & \Sigma_{M_1M_n} \\ \Sigma_{M_2R} & \Sigma_{M_2M_1} & \Sigma_{M_2} & \cdots & \Sigma_{M_2M_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma_{M_nR} & \Sigma_{M_nM_1} & \Sigma_{M_nM_2} & \cdots & \Sigma_{M_n} \end{bmatrix}$$

EKF SLAM

State representation

- Map with n landmarks – $(3 + 2n)$ -dimensional Gaussian

$$\underbrace{\begin{bmatrix} x \\ y \\ \theta \\ m_{1,x} \\ m_{1,y} \\ \vdots \\ m_{n,x} \\ m_{n,y} \end{bmatrix}}_{\mu} \quad \underbrace{\begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{x\theta} & \sigma_{xm_{1,x}} & \sigma_{xm_{1,y}} & \dots & \sigma_{xm_{n,x}} & \sigma_{xm_{n,y}} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{y\theta} & \sigma_{ym_{1,x}} & \sigma_{ym_{1,y}} & \dots & \sigma_{ym_{n,x}} & \sigma_{ym_{n,y}} \\ \sigma_{\theta x} & \sigma_{\theta y} & \sigma_{\theta\theta} & \sigma_{\theta m_{1,x}} & \sigma_{\theta m_{1,y}} & \dots & \sigma_{\theta m_{n,x}} & \sigma_{\theta m_{n,y}} \\ \sigma_{m_{1,x}x} & \sigma_{m_{1,x}y} & \sigma_{m_{1,x}\theta} & \sigma_{m_{1,x}m_{1,x}} & \sigma_{m_{1,x}m_{1,y}} & \dots & \sigma_{m_{1,x}m_{n,x}} & \sigma_{m_{1,x}m_{n,y}} \\ \sigma_{m_{1,y}x} & \sigma_{m_{1,y}y} & \sigma_{m_{1,y}\theta} & \sigma_{m_{1,y}m_{1,x}} & \sigma_{m_{1,y}m_{1,y}} & \dots & \sigma_{m_{1,y}m_{n,x}} & \sigma_{m_{1,y}m_{n,y}} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \sigma_{m_{n,x}x} & \sigma_{m_{n,x}y} & \sigma_{m_{n,x}\theta} & \sigma_{m_{n,x}m_{1,x}} & \sigma_{m_{n,x}m_{1,y}} & \dots & \sigma_{m_{n,x}m_{n,x}} & \sigma_{m_{n,x}m_{n,y}} \\ \sigma_{m_{n,y}x} & \sigma_{m_{n,y}y} & \sigma_{m_{n,y}\theta} & \sigma_{m_{n,y}m_{1,x}} & \sigma_{m_{n,y}m_{1,y}} & \dots & \sigma_{m_{n,y}m_{n,x}} & \sigma_{m_{n,y}m_{n,y}} \end{bmatrix}}_{\Sigma}$$

- Can handle hundreds of dimensions

EKF SLAM

Filter Cycle

1. State prediction (odometry)
2. Measurement prediction
3. Measurement
4. Data association
5. Update
6. Integration of new landmarks

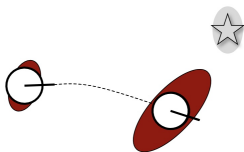
EKF SLAM

Filter Cycle

1. State prediction (odometry)
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EKF SLAM

State Prediction



Odometry

$$\hat{x}_R = f(x_R, u)$$

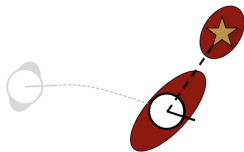
$$\hat{\Sigma}_R = F_x \Sigma_R F_x^T + F_u U F_u^T$$

Robot-landmark cross-covariance prediction $\hat{\Sigma}_{RM_i} = F_x \Sigma_{RM_i}$

$$\mu \rightarrow \begin{bmatrix} x_R \\ m_1 \\ m_2 \\ \vdots \\ m_n \end{bmatrix} \quad \Sigma \rightarrow \begin{bmatrix} \Sigma_R & \Sigma_{RM_1} & \Sigma_{RM_2} & \cdots & \Sigma_{RM_n} \\ \Sigma_{M_1R} & \Sigma_{M_1} & \Sigma_{M_1M_2} & \cdots & \Sigma_{M_1M_n} \\ \Sigma_{M_2R} & \Sigma_{M_2M_1} & \Sigma_{M_2} & \cdots & \Sigma_{M_2M_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma_{M_nR} & \Sigma_{M_nM_1} & \Sigma_{M_nM_2} & \cdots & \Sigma_{M_n} \end{bmatrix}$$

EKF SLAM

Measurement Prediction



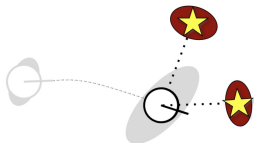
Global-to-local frame transform h

$$\hat{z}_k = h(\hat{x}_k)$$

$$\mu \rightarrow \begin{bmatrix} x_R \\ m_1 \\ m_2 \\ \vdots \\ m_n \end{bmatrix} \quad \Sigma \rightarrow \begin{bmatrix} \Sigma_R & \Sigma_{RM_1} & \Sigma_{RM_2} & \cdots & \Sigma_{RM_n} \\ \Sigma_{M_1R} & \Sigma_{M_1} & \Sigma_{M_1M_2} & \cdots & \Sigma_{M_1M_n} \\ \Sigma_{M_2R} & \Sigma_{M_2M_1} & \Sigma_{M_2} & \cdots & \Sigma_{M_2M_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma_{M_nR} & \Sigma_{M_nM_1} & \Sigma_{M_nM_2} & \cdots & \Sigma_{M_n} \end{bmatrix}$$

EKF SLAM

Obtained Measurement



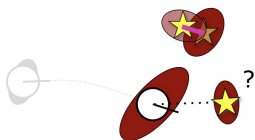
(x, y) -point landmarks

$$z_k = \begin{bmatrix} x_1 \\ y_1 \\ x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}, \quad R_k = \begin{bmatrix} R_1 & 0 \\ 0 & R_2 \end{bmatrix}$$

$$\mu \rightarrow \begin{bmatrix} x_R \\ m_1 \\ m_2 \\ \vdots \\ m_n \end{bmatrix} \quad \Sigma \rightarrow \begin{bmatrix} \Sigma_R & \Sigma_{RM_1} & \Sigma_{RM_2} & \cdots & \Sigma_{RM_n} \\ \Sigma_{M_1R} & \Sigma_{M_1} & \Sigma_{M_1M_2} & \cdots & \Sigma_{M_1M_n} \\ \Sigma_{M_2R} & \Sigma_{M_2M_1} & \Sigma_{M_2} & \cdots & \Sigma_{M_2M_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma_{M_nR} & \Sigma_{M_nM_1} & \Sigma_{M_nM_2} & \cdots & \Sigma_{M_n} \end{bmatrix}$$

EKF SLAM

Data Association



Associates predicted measurements $\hat{z}^{ki}$
with observation z_k^j

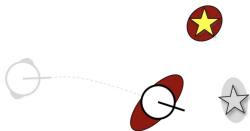
$$v_k^{ij} = z_k^j - \hat{z}_k^i$$

$$S_k^{ij} = R_k^i + H^i \hat{\Sigma}_k H^{iT}$$

$$\mu \rightarrow \begin{bmatrix} x_R \\ m_1 \\ m_2 \\ \vdots \\ m_n \end{bmatrix} \quad \Sigma \rightarrow \begin{bmatrix} \Sigma_R & \Sigma_{RM_1} & \Sigma_{RM_2} & \cdots & \Sigma_{RM_n} \\ \Sigma_{M_1R} & \Sigma_{M_1} & \Sigma_{M_1M_2} & \cdots & \Sigma_{M_1M_n} \\ \Sigma_{M_2R} & \Sigma_{M_2M_1} & \Sigma_{M_2} & \cdots & \Sigma_{M_2M_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma_{M_nR} & \Sigma_{M_nM_1} & \Sigma_{M_nM_2} & \cdots & \Sigma_{M_n} \end{bmatrix}$$

EKF SLAM

Update Step



The usual Kalman filter expressions

$$K_k = \hat{\Sigma}_k H^T S_k^{-1}$$

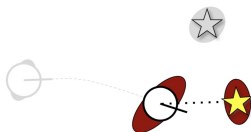
$$x_k = \hat{x}_k + K_k v_k$$

$$C_k = (I - K_k H) \hat{\Sigma}_k$$

$$\mu \rightarrow \begin{bmatrix} x_R \\ m_1 \\ m_2 \\ \vdots \\ m_n \end{bmatrix} \quad \Sigma \rightarrow \begin{bmatrix} \Sigma_R & \Sigma_{RM_1} & \Sigma_{RM_2} & \cdots & \Sigma_{RM_n} \\ \Sigma_{M_1 R} & \Sigma_{M_1} & \Sigma_{M_1 M_2} & \cdots & \Sigma_{M_1 M_n} \\ \Sigma_{M_2 R} & \Sigma_{M_2 M_1} & \Sigma_{M_2} & \cdots & \Sigma_{M_2 M_n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Sigma_{M_n R} & \Sigma_{M_n M_1} & \Sigma_{M_n M_2} & \cdots & \Sigma_{M_n} \end{bmatrix}$$

EKF SLAM

New Landmarks



State augmented by

$$m_{n+1} = g(x_R, z_j)$$

$$\Sigma_{M_{n+1}} = G_R \Sigma_R G_R^T + G_z R_j G_z^T$$

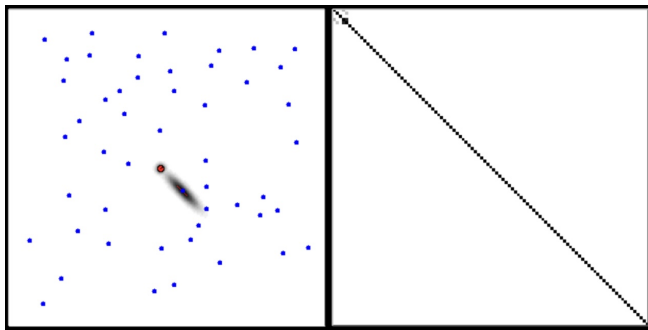
Cross-covariances

$$\Sigma_{M_{n+1}M_i} = G_R \Sigma_{RM_i}$$

$$\Sigma_{M_{n+1}R} = G_R \Sigma_R$$

$$\begin{bmatrix} x_R \\ m_1 \\ m_2 \\ \vdots \\ m_n \\ m_{n+1} \end{bmatrix} \quad \Sigma \rightarrow \quad \begin{bmatrix} \Sigma_R & \Sigma_{RM_1} & \Sigma_{RM_2} & \cdots & \Sigma_{RM_n} & \Sigma_{RM_{n+1}} \\ \Sigma_{M_1R} & \Sigma_{M_1} & \Sigma_{M_1M_2} & \cdots & \Sigma_{M_1M_n} & \Sigma_{M_1M_{n+1}} \\ \Sigma_{M_2R} & \Sigma_{M_2M_1} & \Sigma_{M_2} & \cdots & \Sigma_{M_2M_n} & \Sigma_{M_2M_{n+1}} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \Sigma_{M_nR} & \Sigma_{M_nM_1} & \Sigma_{M_nM_2} & \cdots & \Sigma_{M_n} & \Sigma_{M_nM_{n+1}} \\ \Sigma_{M_{n+1}R} & \Sigma_{M_{n+1}M_1} & \Sigma_{M_{n+1}M_2} & \Sigma_{M_{n+1}M_n} & \cdots & \Sigma_{M_{n+1}} \end{bmatrix}$$

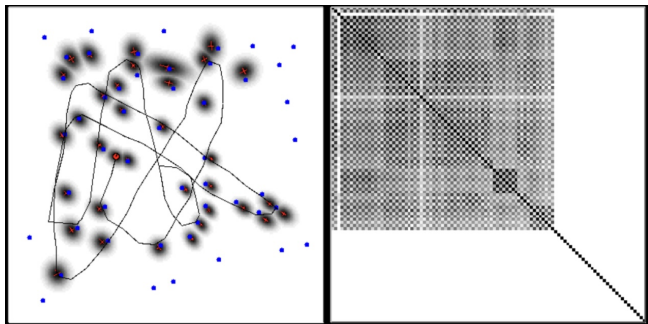
EKF SLAM



Map

Correlation matrix

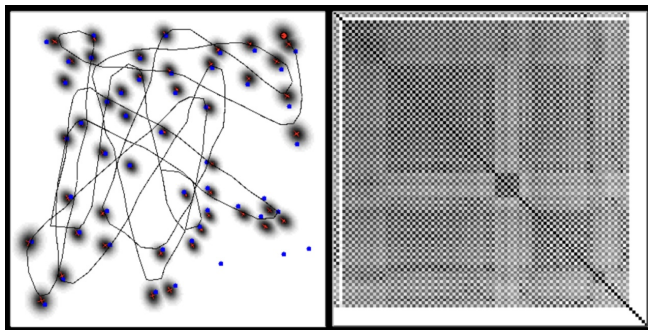
EKF SLAM



Map

Correlation matrix

EKF SLAM



Map

Correlation matrix

EKF SLAM

Correlations Matter

- What if we neglected cross-correlations?

$$\Sigma_k = \begin{bmatrix} \Sigma_R & 0 & \dots & 0 \\ 0 & \Sigma_{M-1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \Sigma_{M_n} \end{bmatrix} \quad \Sigma_{RM_i} = 0_{3 \times 2}, \quad \Sigma_{M_i M_{i+1}} = 0_{2 \times 2}$$

EKF SLAM

Correlations Matter

- What if we neglected cross-correlations?

$$\Sigma_k = \begin{bmatrix} \Sigma_R & 0 & \dots & 0 \\ 0 & \Sigma_{M-1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \Sigma_{M_n} \end{bmatrix} \quad \Sigma_{RM_i} = 0_{3 \times 2}, \quad \Sigma_{M_i M_{i+1}} = 0_{2 \times 2}$$

- Landmark and robot uncertainties would become overly optimistic
- Data association would fail
- Multiple map entries of the same landmark
- Inconsistent map

SLAM

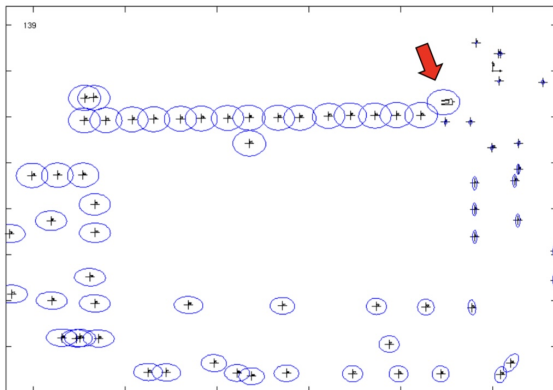
Loop Closure

- Recognizing an already mapped area, typically after a long exploration path (the robot “closes a loop”)
- Structurally identical to data association, but
 - high levels of ambiguity
 - possibly useless validation gates
 - environment symmetries
- Uncertainties collapse after a loop closure (whether the closure was correct or not)

SLAM

Loop Closure

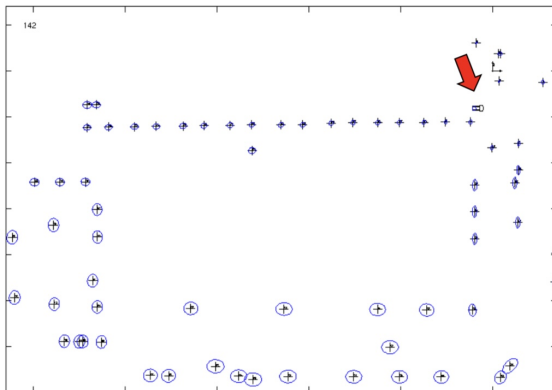
■ Before loop closure



SLAM

Loop Closure

■ After loop closure



SLAM

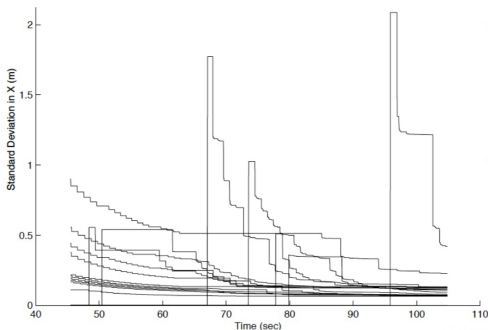
Loop Closure

- By revisiting already mapped areas, uncertainties in robot and landmark estimates can be **reduced**
- This can be exploited when **exploring** an environment for the sake of better (e.g. more accurate) maps
- Exploration: the problem of **where to acquire new information**

KF-SLAM Properties

Linear Case

- The **determinant** of any sub-matrix of the map covariance matrix **decreases monotonically** as successive observations are made

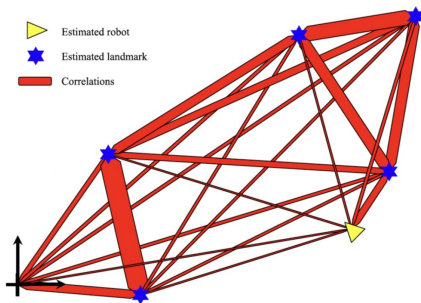


- When a new landmark is initialized, its **uncertainty** is **maximal**
- Landmark uncertainty **decreases monotonically** with each new observation

KF-SLAM Properties

Linear Case

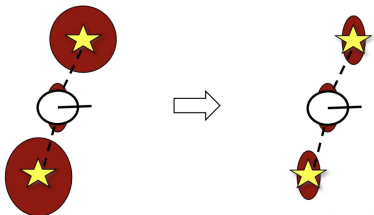
- In the limit, the landmark estimates become **fully correlated**



KF-SLAM Properties

Linear Case

- In the limit, the **covariance** associated with any single landmark location estimate is determined only by the **initial covariance in the vehicle location estimate**



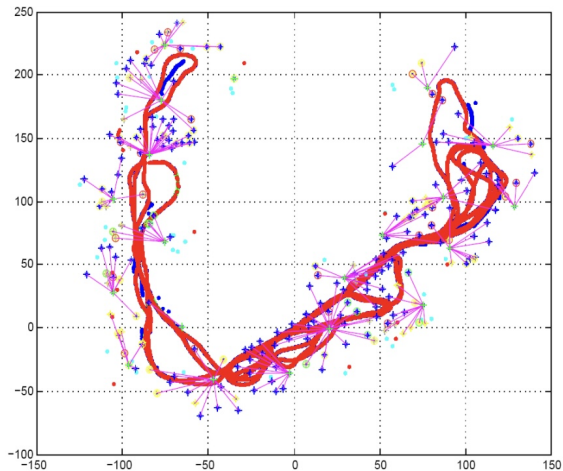
EKF SLAM Example

Victoria Park Dataset



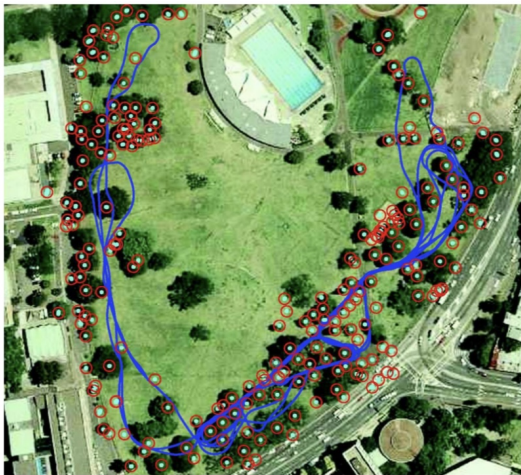
Victoria Park

Estimated Trajectory



Victoria Park

Landmarks



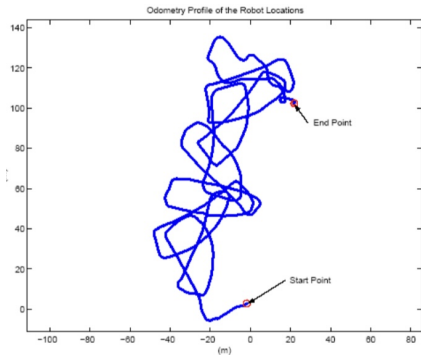
EKF SLAM Example

Tennis Court

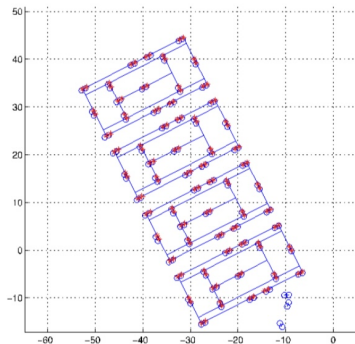


EKF SLAM Example

Tennis Court



Odometry



Estimated trajectory

EKF SLAM

Complexity

- Cost per step – quadratic in n , the number of landmarks – $O(n^2)$
- Total cost to build a map with n landmarks – $O(n^3)$
- Memory consumption – $O(n^2)$
- Problem – becomes computationally intractable for large maps!
- There exists variants to circumvent these problems

SLAM Techniques

- EKF SLAM
- FastSLAM
- Graph-based SLAM
- Topological SLAM (mainly place recognition)
- Scan Matching / Visual Odometry (only locally consistent maps)
- Approximations for SLAM – Local submaps, Sparse extended information filters, Sparse links, Thin junction tree filters, etc.

EKF SLAM

Summary

- The first SLAM solution
- Convergence proof for linear Gaussian case
- Can diverge if nonlinearities are large (and the real world is nonlinear...)
- Can deal only with a single mode
- Successful in medium-scale scenes
- Approximations exists to reduce the computational complexity